Symmetric parity predictors require large amplitudes

An exponential lower bound and a consequence for margin repairs

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Abstract

A symmetric polynomial on the Boolean cube that predicts parity with error at most 1/3, has absolute value at least one everywhere, and has degree at most a sufficiently small fixed fraction of the dimension must have L^1 norm at least $\exp(c\sqrt{k})$. The allowed degree fraction can be any fixed $\alpha < 1 - \sqrt{3}/2$. The proof uses a binomial generating function, an elementary Schrödinger barrier estimate, and an averaged form of Jensen's formula. It includes arbitrary choices of sign pattern and polynomial amplitudes. We also prove that every such symmetric head has diverging average cost in an explicitly defined cusp model of margin repairs, even when repairs may change unobserved signs at intermediate radii. This cost conclusion also allows nonsymmetric amplitudes when the head's sign pattern is symmetric in a sufficiently large block and its degree in that block obeys the stated fractional bound. Sign symmetry and eventual agreement with the head remain explicit hypotheses; no reduction of a general Bayesian posterior to this structured problem is asserted.

1 Statement and scope

Write $\chi_k(x) = \prod_{i=1}^k x_i$ for parity on $\{-1,1\}^k$, equipped with the uniform measure. A symmetric real polynomial P depends only on the Hamming weight $H = |\{i : x_i = -1\}|$. We use $P(h)$ for its value on that layer and

$$w_k(h) = 2^{-k}\binom{k}{h}, \qquad \|P\|_p = \left(\sum_{h=0}^{k} w_k(h)|P(h)|^p\right)^{1/p}.$$

Polynomial degree means multilinear Fourier degree on the cube. For a symmetric function this is also the degree in its Krawtchouk expansion. Its parity advantage is

$$c(P) = \mathbb{E}[\chi_k \operatorname{sign} P] = 1 - 2\mathbb{P}(\operatorname{sign} P \neq \chi_k).$$

All logarithms below are natural. The first four sections prove the amplitude bound on the finite cube. The final section introduces a separate repair-energy model and states the additional anchoring hypothesis needed to transfer that bound to repair costs.

Theorem 1 (Exponential amplitude lower bound)**.** *Fix $0 < \rho < 1$ and a degree fraction α satisfying*

$$0 < \alpha < 1 - \cos(\pi\rho/2). \tag{1}$$

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There are constants $c > 0$ and k_0, depending only on α, ρ, with the following property. If $k \geq k_0$ and P is symmetric with

$$\deg P \leq \alpha k, \qquad |P(x)| \geq 1 \text{ for every } x, \qquad c(P) \geq \rho,$$

then $\|P\|_1 \geq \exp(c\sqrt{k})$.

For error at most $1/3$, take $\rho = 1/3$. The allowed degree fraction is any fixed $\alpha < 1 - \sqrt{3}/2 = 0.1339745962\ldots$. The constants and the threshold are not claimed optimal.

The theorem concerns every symmetric polynomial of the indicated fractional degree, not merely a particular interpolation polynomial or a prescribed sign pattern. The norm conclusion is stronger than any polynomial lower bound in k. It does not hold for unrestricted nonsymmetric heads: substituting growing disjoint block parities into a fixed accurate polynomial preserves its norm and relative degree. Such an outer polynomial can have arbitrarily small relative degree: interpolate parity on a sufficiently wide central $O(\sqrt{m})$ window of an m-cube and rescale to margin one, then hold m fixed while increasing the substituted block sizes. Accordingly, this note supplies no symmetrization theorem and no sample-complexity theorem for a general posterior.

2 A Fourier cutoff from the degree bound

The key estimate uses the concentration of a finite-dimensional Schrödinger spectral subspace. We give the needed estimate directly.

Lemma 2 (Weighted Fourier concentration)**.** *Fix $0 < \alpha < 1$ and $\theta \in (0, \pi)$ with $\alpha < 1 - \cos(\theta/2)$. There are $\kappa > 0$ and k_0 such that, for every $k \geq k_0$ and symmetric polynomial P of degree $r \leq \alpha k$, the quantities*

$$a_h = w_k(h)P(h), \qquad Q(t) = \sum_{h=0}^{k} a_h e^{ith}, \qquad M = \|P\|_1$$

satisfy

$$\sup_{\theta \leq |t| \leq \pi} |Q(t)| \leq M e^{-\kappa k}. \tag{2}$$

No margin assumption is needed.

Proof. Let $\mathcal{Q}(z) = \sum_h w_k(h)P(h)z^h$, so that $Q(t) = \mathcal{Q}(e^{it})$. The identity

$$\sum_{h=0}^{k} \binom{k}{h} (h)_j z^h = (k)_j z^j (1+z)^{k-j}$$

for falling factorials shows that $\mathcal{Q}(z) = (1+z)^{k-r}R(z)$, where R is real and has degree at most r. Put $m = k - r$ and $x = t/2$. Then

$$e^{-ikx}\mathcal{Q}(e^{2ix}) = 2^m \cos^m x \, e^{-irx} R(e^{2ix}).$$

The last factor has frequencies $r, r-2, \ldots, -r$. Elementary trigonometric identities imply that the real and imaginary parts of the centered generating function, in one order according to the parity of r, belong to the spaces

$$V_{m,r} = \{\cos^m x \, p(\sin x) : \deg p \leq r\}, \qquad V_{m+1,r-1}.$$

The second component is zero when $r = 0$. The constant 2^m is absorbed into the coefficients. Each real component has ordinary $L^2(-\pi/2, \pi/2)$ norm at most $\sqrt{\pi}M$, because $|Q(t)| \le M$.

For $\lambda \ge 2$, let p_n be orthonormal polynomials for the weight $(1-y^2)^{\lambda-1/2}\,dy$ on $(-1,1)$. Integration by parts shows that

$$L_\lambda p = -(1-y^2)p'' + (2\lambda+1)yp'$$

is symmetric for this weight. It preserves degree and has leading coefficient $n(n+2\lambda)$ on a polynomial of degree n. Orthogonality therefore gives $L_\lambda p_n = n(n+2\lambda)p_n$. The functions $\phi_n(x) = \cos^\lambda x\, p_n(\sin x)$ are orthonormal in dx, span $V_{\lambda,s}$ for $0 \le n \le s$, and satisfy, by direct differentiation,

$$-\phi_n'' + \lambda(\lambda-1)\sec^2 x\,\phi_n = (n+\lambda)^2\phi_n. \tag{3}$$

In both spaces above, $\lambda \ge m$ and $n + \lambda \le k$.

Choose $\arccos(1-\alpha) < x_1 < x_0 = \theta/2$. For some $\delta > 0$ and all large k, the potential $V = \lambda(\lambda-1)\sec^2 x$ and energy $E = (n+\lambda)^2$ obey

$$V - E \ge m(m-1)\sec^2 x - k^2 \ge \delta k^2 \qquad (|x| \ge x_1).$$

Set $F(x) = k\sqrt{\delta/2}(|x| - x_1)_+$. Multiplying (3) by $e^{2F}\phi_n$ and integrating gives

$$\int \Big(|(e^F\phi_n)'|^2 + (V - E - |F'|^2)e^{2F}|\phi_n|^2 \Big)\,dx = 0. \tag{4}$$

Boundary terms vanish since $\lambda \ge 2$ for large k. The Lipschitz weight is justified directly in the weak identity, or by smoothing. Outside $[-x_1, x_1]$ the second coefficient is at least $\delta k^2/2$; inside, $F = 0$ and $V - E \ge -k^2$. Consequently

$$\|(e^F\phi_n)'\|_2 \le k, \qquad \|e^F\phi_n\|_2^2 \le 1 + 2/\delta.$$

The elementary one-dimensional bound $|u(x)|^2 \le 2\|u\|_2\|u'\|_2$ for a function vanishing at the endpoints now implies

$$|\phi_n(x)| \le C\sqrt{k}\,e^{-ck} \qquad (|x| \ge x_0),$$

where $c = \sqrt{\delta/2}(x_0 - x_1) > 0$. Expand either real component in its orthonormal basis and use Cauchy–Schwarz. There are at most $k+1$ terms, so each component is bounded by Cke^{-ck} times its L^2 norm in this region. The norm bound $\sqrt{\pi}M$ proves (2), after decreasing c to absorb the polynomial factor and increasing k_0. □

The low-pass interpolation

$$F(z) = \frac{1}{2\pi}\int_{-\theta}^{\theta} Q(t)e^{-itz}\,dt \tag{5}$$

is entire and real on the real axis. Since $|Q(t)| \le M$, it satisfies

$$|F(z)| \le Me^{\theta|\operatorname{Im} z|}, \qquad |F(h) - a_h| \le Me^{-\kappa k} \quad (h \in \mathbb{Z}). \tag{6}$$

The second inequality follows by subtracting the omitted part of the Fourier inversion formula. A hard frequency cutoff suffices.

3 Parity advantage forces sign transitions

Lemma 3. *For any $g_0, \dots, g_k \in \{-1, 1\}$,*

$$\left| \sum_{h=0}^{k} (-1)^h w_k(h) g_h \right| \le \sum_{h : g_h \neq g_{h+1}} w_{k-1}(h). \tag{7}$$

Proof. The alternating cumulative-binomial identity is $\sum_{i=0}^{h} (-1)^i \binom{k}{i} = (-1)^h \binom{k-1}{h}$. Summation by parts therefore expresses the sum on the left as

$$-2^{-k} \sum_{h=0}^{k-1} (g_{h+1} - g_h)(-1)^h \binom{k-1}{h}.$$

Taking absolute values proves the result. □

We need a simple discrete logarithmic kernel. For $R > 1$ define

$$L_R(j) = \left[\log \frac{R}{|j| + 1} \right]_+, \qquad S_R = \sum_{j \in \mathbb{Z}} L_R(j).$$

Stirling's formula, or comparison with an integral, gives

$$S_R = 2R + O(\log R). \tag{8}$$

If $F(j)$ and $F(j+1)$ have opposite signs, a real zero lies between them. In Jensen's formula centered at an integer h, that zero contributes at least $L_R(j - h)$ at radius R.

4 Averaging Jensen's formula

Proof of Theorem 1. By (1), choose $2 \arccos(1 - \alpha) < \theta < \pi\rho$. Use F and κ from (5)–(6). If $\log M > \kappa k/2$, the conclusion is already true for large k. We may therefore assume $M \le e^{\kappa k/2}$.

Fix a large constant B and then a small constant $\varepsilon > 0$, and put $R = \varepsilon\sqrt{k}$. Throughout the central window $|h - k/2| \le (B + 2\varepsilon)\sqrt{k}$, binomial weights are bounded below by $c_{B,\varepsilon}/\sqrt{k}$. Thus (6) and the margin assumption imply, for all large k,

$$\operatorname{sign} F(h) = \operatorname{sign} P(h), \qquad |F(h)| \ge \tfrac{1}{2} w_k(h). \tag{9}$$

Let J be the set of sign transitions j with $|j - k/2| \le B\sqrt{k}$. Lemma 3 and a binomial tail bound imply

$$\sum_{j \in J} w_k(j) \ge \rho - \tau_B - o(1), \qquad \tau_B \longrightarrow 0 \quad (B \longrightarrow \infty). \tag{10}$$

Here $w_{k-1}(j)/w_k(j) = 2(k - j)/k = 1 + o(1)$ uniformly in the fixed central window. For example one can take $\tau_B = 2e^{-2(B - o(1))^2}$ before absorbing the vanishing term.

For central j and $|h - j| \le R$, adjacent binomial ratios give

$$w_k(h) \ge \eta_{B,\varepsilon,k} w_k(j), \qquad \liminf_{k \to \infty} \eta_{B,\varepsilon,k} \ge e^{-C\varepsilon(B+\varepsilon)}. \tag{11}$$

Indeed each adjacent log ratio is $O((B + \varepsilon)/\sqrt{k})$ in this window, and there are at most $\varepsilon\sqrt{k}$ such steps.

Apply Jensen's formula to F centered at each integer h in the anchor window $I = \{|h - k/2| \le (B + 2\varepsilon)\sqrt{k}\}$. Keeping only the zeros supplied by J, and using (6), gives

$$\sum_{j \in J} L_R(j - h) \le \log M - \log |F(h)| + \frac{2\theta R}{\pi}.$$

Zeros on the integration circle are handled by a limiting radius. Multiply by $w_k(h)$ and sum over $h \in I$. The right side is at most

$$\log M + O_{B,\varepsilon}(\log k) + \frac{2\theta R}{\pi}$$

by (9). On the left, (11) and (8) give at least

$$\eta_{B,\varepsilon,k} S_R \sum_{j \in J} w_k(j).$$

Every nonzero kernel term has its anchor in I, so no term is missing. Combining the estimates yields

$$\log M \ge 2R \left[\eta_{B,\varepsilon,k}(\rho - \tau_B - o(1)) - \frac{\theta}{\pi} \right] - O_{B,\varepsilon}(\log k). \tag{12}$$

Choose B so that τ_B is much smaller than $\rho - \theta/\pi$, then choose ε so that the loss in (11) is also smaller than this gap. The bracket is then bounded below by a positive constant. Since $R = \varepsilon\sqrt{k}$, the theorem follows. □

Proposition 4 (Vanishing margin defects)**.** *Under the strict parameter condition of Theorem 1, its conclusion remains valid along sequences of symmetric polynomials P_k with $\deg P_k \le \alpha k$ for which, for some fixed $a > 0$,*

$$\mathbb{P}(|P_k| < a) = o(1), \qquad c(P_k) \ge \rho - o(1).$$

More precisely, $\liminf k^{-1/2} \log \|P_k\|_1 > 0$.

Proof. Use the same central windows and remove transitions incident to a vertex with $|P_k| < a$. Their total central binomial weight is $o(1)$, so (10) changes by $o(1)$. At the remaining endpoints the low-pass interpolation preserves signs. Also omit bad anchor vertices from the averaged Jensen inequality. The lost left-hand contribution is at most

$$\sum_{h \text{ bad}} w_k(h) \sum_{j \in \mathbb{Z}} L_R(j - h) = S_R \mathbb{P}(h \text{ bad}) = o(R).$$

At every retained anchor, $|F(h)| \ge a w_k(h)/2$. Thus (12) changes only by $o(R) + O_a(1)$, proving the claim. □

If a symmetric P_k of degree at most αk is $o(1)$ in L^2 from a globally margin-one function Z_k of parity error at most $(1 - \rho)/2$, then the proposition applies with $a = 1/2$. The approximating function Z_k itself need not be symmetric.

Corollary 5 (Symmetry in one block)**.** *Let P be a globally margin-one polynomial on k target coordinates, invariant under permutations of a block B of n coordinates. Suppose its degree in the coordinates of B is at most αn and its global parity advantage is at least ρ, with the strict parameter condition of Theorem 1. Arbitrary dependence on the other coordinates is allowed. For all sufficiently large n,*

$$\|P\|_1 \ge c \exp(c'\sqrt{n}),$$

where the positive constants depend only on α, ρ.

Proof. Choose $0 < \rho_0 < \rho$ close enough that the strict parameter condition still holds. Condition on all coordinates outside B, and absorb their parity into the sign of the resulting symmetric polynomial. The conditional parity advantages lie in $[-1, 1]$ and have mean at least ρ. Thus a fraction at least $(\rho - \rho_0)/(1 - \rho_0)$ of fibers have advantage at least ρ_0. Each such fiber has the same degree and margin bounds, so Theorem 1 applies uniformly. Average its conclusion over these fibers. □

5 A consequence for all-radius margin repairs

We now define an auxiliary geometric problem independently of any neural-network posterior. Fix $0 < \beta \le 1/2$ and let the active points be $u \in \{-1, 1\}^k$, with an integer radius $h \ge 0$. For any finitely supported real charge $q = (q_{u,h})$ of total mass zero, put

$$\mathcal{V}_\beta(q) = -\sum_{u,h,v,j} q_{u,h} q_{v,j} (d_H(u, v) + h + j)^\beta. \tag{13}$$

This quadratic form is nonnegative: use the integral representation of t^β and positivity of the kernels $e^{-td_H(u,v)} e^{-th} e^{-tj}$. Define the energy of a repair g by

$$\mathcal{E}_\beta(g) = \sup_{q:\, \sum q = 0,\ \mathcal{V}_\beta(q) > 0} \frac{\left| \sum_{u,h} q_{u,h} g(u, h) \right|^2}{\mathcal{V}_\beta(q)},$$

with infinite energy if a zero-variance charge has nonzero pairing. This definition specifies the only norm property used below.

A head P has $|P(u)| \ge 1$. At each wrong point u, a repair g_u is admissible if

$$|P(v) + g_u(v, h)| \ge 1 \quad \text{for all } v, h, \qquad \operatorname{sign}(P(u) + g_u(u, 0)) = \chi_k(u),$$

and $g_u(v, h) \to 0$ as $h \to \infty$ for every v. Intermediate signs are unrestricted. The last condition supplies eventual agreement with the original head and is an explicit hypothesis of this model.

Lemma 6 (Contact drop with arbitrary intermediate sign changes)**.** *Suppose $P(u)$ and $P(v)$ have the same sign, and $D = 2 + |P(u)| - |P(v)| > 0$. Every admissible repair at u has energy E satisfying*

$$E \ge c_\beta \min\left\{ \frac{D^2}{(d_H(u, v) + 1)^\beta},\ D^{2-\beta} \right\}.$$

Proof. The charge comparing (v, h) and $(v, h + 1)$ has variance

$$v_h = 2(2h + 1)^\beta - (2h)^\beta - (2h + 2)^\beta \le C_\beta (h + 1)^{\beta - 2}.$$

A sign change between these radii requires a jump of at least 2, and therefore energy at least $4/v_h$. Choose an integer $R \le C_\beta(1 + E^{1/(2-\beta)})$ large enough that this lower bound exceeds E for every $h \ge R$. Eventual agreement implies agreement already at radius R. Comparing $(u, 0)$ with (v, R) now gives a repair difference of absolute value at least D. Its variance is

$$2(d_H(u, v) + R)^\beta - (2R)^\beta \le C_\beta [(d_H(u, v) + 1)^\beta + E^{\beta/(2-\beta)}].$$

Thus D^2 is bounded by $C_\beta E$ times the last bracket. At least one of its two terms accounts for half of D^2, proving the result. □

Theorem 7 (Symmetric heads have diverging population repair cost)**.** *Fix $\beta \in (0, 1/2]$ and a degree fraction α satisfying (1) with $\rho = 1/3$. For all sufficiently large k, every symmetric head P with degree at most αk, global margin one, and parity error at most $1/3$, and every family of admissible repairs, satisfy*

$$\mathbb{E}_u \left[\mathbf{1}_{\{\operatorname{sign} P(u) \neq \chi_k(u)\}} \mathcal{E}_\beta(g_u) \right] \geq \frac{c_\beta}{(k+1)^\beta} \exp(c\sqrt{k}),$$

where $c > 0$ may depend on α, β.

Proof. Both signs occur, since a constant sign has parity error $1/2$. Let b_+ and b_- be the smallest positive and negative absolute head values, and define

$$s = (b_+ + b_-)/2 \geq 1, \qquad t = (b_+ - b_-)/2, \qquad Q = (P - t)/s.$$

Then Q is symmetric, has the same signs and degree bound, satisfies $|Q| \geq 1$, and has a contact at each of $+1$ and -1. For every wrong u, choose a same-sign contact v. The contact drop for the original head is

$$D = 2 + |P(u)| - b_{\operatorname{sign} P(u)} = 2 + s(|Q(u)| - 1) \geq 1 + |Q(u)|.$$

Lemma 6, using $d_H(u, v) \leq k$, gives, with $p = 2 - \beta$,

$$\mathcal{E}_\beta(g_u) \geq \frac{c_\beta}{(k+1)^\beta} |Q(u)|^p.$$

Since $\deg Q < k$, parity orthogonality gives $\mathbb{E}[\chi_k Q] = 0$. Consequently the wrong and correct sides have equal absolute first moments, and

$$\mathbb{E}[\mathbf{1}_{\text{wrong}} |Q|] = \tfrac{1}{2} \|Q\|_1.$$

Hölder's inequality implies $\mathbb{E}[\mathbf{1}_{\text{wrong}} |Q|^p] \geq 2^{-p} \|Q\|_1^p$. Apply Theorem 1 to Q and average the preceding energy bound. □

The same proof applies to a head symmetric in a block of n target coordinates, with block degree at most αn: replace the final application of Theorem 1 by Corollary 5. Parity orthogonality still holds because the degree in that block is less than n, regardless of the degree in the other coordinates. The resulting lower bound is $c(k+1)^{-\beta} \exp(c'\sqrt{n})$. In particular it diverges exponentially in $\sqrt{k}$ when the block contains a fixed positive fraction of the target coordinates.

Corollary 8 (Only the sign pattern needs block symmetry)**.** *Let P have global margin one on k target coordinates, and suppose $\operatorname{sign} P$ is invariant under permutations of a block of m coordinates. The amplitudes and the dependence on coordinates outside that block may be arbitrary. If the block degree is at most αm, the parity error is at most 0.34, and*

$$0 < \alpha < 1 - \cos(4\pi/25) = 0.123693319956\ldots,$$

then every family of admissible repairs satisfies, for sufficiently large m,

$$\mathbb{E}_u[\mathbf{1}_{\text{wrong}} \mathcal{E}_\beta(g_u)] \geq c(m+1)^{-\beta} \exp(c'\sqrt{m}).$$

The constants may depend on the strict gap in this degree condition.

Proof. Choose $0 < \rho_0 < 0.32$ sufficiently close to 0.32 that $\alpha < 1 - \cos(\pi\rho_0/2)$. Condition on coordinates outside the block. Since the mean conditional parity advantage is at least 0.32, a fraction at least $(0.32-\rho_0)/(1-\rho_0) > 0$ of the fibers have advantage at least ρ_0. Fix such a fiber and absorb the fixed outside parity into its target f. Write $s = \operatorname{sign} P$, $E = \{fs = -1\}$, $C = \{fs = 1\}$, $R = \mathbb{P}(E) \le (1-\rho_0)/2$, and $p = 2 - \beta$. Define

$$\Phi_s(P) = \mathbb{E}_u \mathbf{1}_E(u) \max_{v \in C:\, s(v)=s(u)} \frac{[s(u)(P(u) - P(v))]_+^p}{(1 + d_H(u,v))^\beta}.$$

This functional is convex in P and invariant under permutations of the block. Averaging P over those permutations gives a symmetric polynomial $\overline{P}$ with the same signs, margin, and degree bound, and $\Phi_s(\overline{P}) \le \Phi_s(P)$. The contact-drop lemma implies that the original repair cost on this fiber is at least $c_\beta \Phi_s(P)$.

For completeness, Φ_s controls the first moment uniformly over this sign cone. Put $c_0 = 1 - 2R$ and $b_\sigma = \min_{v \in C: s(v)=\sigma} |\overline{P}(v)|$. Each phase has correct mass minus wrong mass $c_0/2$. With $D_u = (|\overline{P}(u)| - b_{s(u)})_+$ on E, parity orthogonality gives

$$\frac{c_0}{2}(b_+ + b_-) \le \mathbb{E}[\mathbf{1}_E D_u], \qquad \|\overline{P}\|_1 \le \frac{2}{c_0}\mathbb{E}[\mathbf{1}_E D_u].$$

Consequently Hölder's inequality yields

$$\|\overline{P}\|_1 \le \frac{2}{c_0} R^{1-1/p} (m+1)^{\beta/p} \Phi_s(\overline{P})^{1/p}.$$

Theorem 1, with advantage ρ_0, now gives $\Phi_s(P) \ge c(m+1)^{-\beta} \exp(c'\sqrt{m})$. Averaging over the favorable fibers proves the claim. Restricting contact tests to one fiber is valid for every original repair, so no symmetry of the repairs or of the original amplitudes was imposed. □

This result allows every intermediate sign change and every choice of repair values. Its eventual-agreement condition and the stated symmetry of the head or its sign pattern remain assumptions of the statement. In particular, neither the rotational symmetry of a prior nor the radial form of a kernel makes a head fitted to an individual training set symmetric. A general posterior lower bound requires an additional argument beyond this note.

References

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